RL Circuit: Natural Response

Figure 7.16 - Solution Key

Circuit	40 V source; 2 H inductor
Resistors	2 Ω, 4 Ω, 12 Ω and 16 Ω
Switching event	Switch opens at t = 0
Required result	Inductor current i(t), for t > 0

Problem Statement

The switch has been closed for a long time. At t = 0, it is opened. Determine the current through the 2 H inductor for t > 0, with downward current taken as positive. All circuit elements are assumed ideal.

1. Initial Current Before Opening

At DC steady state, the ideal inductor behaves as a short circuit. This short places zero voltage across the 16 Ω resistor, so that resistor carries no current.

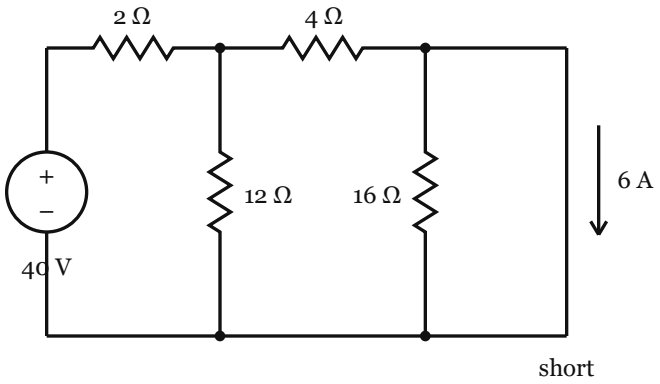


Figure 1. Closed-switch steady-state equivalent; the inductor is a short.

The source sees 2 Ω in series with the parallel pair (12 Ω and 4 Ω):

$$R_{total} = 2 + (12 \times 4)/(12 + 4) = 5\ \Omega$$

$$I_{source} = 40/5 = 8\ A$$

The voltage across the parallel branches is 24 V. The current through the 4 Ω branch flows through the inductor short:

$$V_{parallel} = 40 - 8(2) = 24\ V$$

$$\mathbf{i(0^-) = 24/4 = 6\ A}$$

Inductor current cannot change instantaneously. Therefore, the initial current immediately after opening is:

$$\mathbf{i(0^+) = i(0^-) = 6\ A}$$

2. Resistance After Opening

Opening the switch disconnects the 40 V source and the 2 Ω resistor. Between the inductor terminals A and B, the 4 Ω and 12 Ω resistors form one series path; the 16 Ω resistor forms the other path.

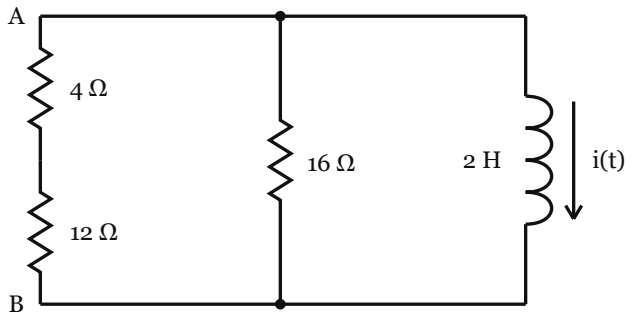


Figure 2. Open-switch circuit, redrawn to show the two resistive paths.

$$R_{eq} = [(4 + 12) \times 16]/[(4 + 12) + 16] = 8\ \Omega$$

3. Time Constant and Natural Response

The remaining circuit contains no independent source. Stored magnetic energy is dissipated in the equivalent resistance, and the current decays toward zero.

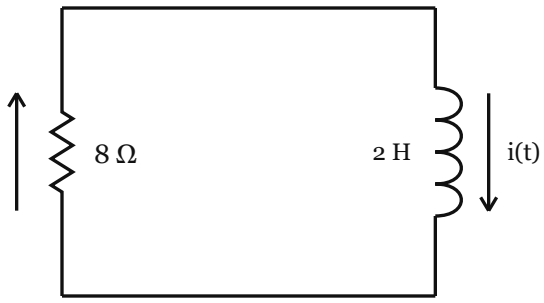


Figure 3. Equivalent 8 Ω resistance and 2 H inductor.

$$\tau = L/R_{eq} = 2/8 = 0.25\ \text{s}$$

$$i(t) = i(0^+)e^{-t/\tau}$$

$$\boxed{\mathbf{i(t) = 6e^{-4t}\ A,\quad t > 0}}$$

Here t is in seconds and the current is downward. As a check, $i(0^+) = 6$ A, $i(0.25\ \text{s}) \approx 2.21$ A, and i(t) approaches zero as t increases.